

Proof. Define R_i to be the indicator random variable for A_i , where $R_i(\omega) = 1$ if $\omega \in A_i$ and $R_i(\omega) = 0$ if $\omega \notin A_i$. Let $R = R_1 + R_2 + \cdots + R_n$. Then

$$\begin{aligned} \text{Ex}[R] &= \sum_{i=1}^n \text{Ex}[R_i] && \text{(by Linearity of Expectation)} \\ &= \sum_{i=1}^n \Pr[R_i = 1] && \text{(by Lemma 18.4.2)} \\ &= \sum_{i=1}^n \Pr[A_i]. && \text{(def of indicator variable)} \end{aligned}$$

So whenever you are asked for the expected number of events that occur, all you have to do is sum the probabilities that each event occurs. Independence is not needed.

18.5.3 Expectation of a Binomial Distribution

Suppose that we independently flip n biased coins, each with probability p of coming up heads. What is the expected number of heads?

Let J be the random variable denoting the number of heads. Then J has a binomial distribution with parameters n , p , and

$$\Pr[J = k] = \binom{n}{k} p^k (1-p)^{n-k}.$$

Applying equation (18.3), this means that

$$\text{Ex}[J] = \sum_{k=0}^n k \Pr[J = k] = \sum_{k=0}^n k \binom{n}{k} p^k (1-p)^{n-k}. \quad (18.11)$$

This sum looks a tad nasty, but linearity of expectation leads to an easy derivation of a simple closed form. We just express J as a sum of indicator random variables, which is easy. Namely, let J_i be the indicator random variable for the i th coin coming up heads, that is,

$$J_i ::= \begin{cases} 1 & \text{if the } i \text{th coin is heads} \\ 0 & \text{if the } i \text{th coin is tails.} \end{cases}$$

Then the number of heads is simply

$$J = J_1 + J_2 + \cdots + J_n.$$